No.	Status	Date
LNDE-RPT-20140902-024-DS-4-RA-101	Issued	2014-09-19

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 4-RA-101 located within ATU-100. The scope of work consists of a pre-change rich-amine survey to establish current wall thickness and evaluate asset integrity prior to MOC implementation.

Marcus Chen, acting as lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the data presented herein.

Technical Basis

All thickness measurements were performed in accordance with QSCP-NDT-001 Rev 2, which incorporates specific requirements for insulation removal, reinstatement, and inspection-port access as part of the CUI program.

The inspection methodology and equipment calibration are based on the requirements of ASTM E797.

Line Specifications

The subject line, 4-RA-101, is constructed from A106 Gr.B with a nominal outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design limits for this circuit are 862 kPa(g) and 82 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. Any reading recorded below this value is subject to an NCR.

Inspection Context

This data sheet supports the findings detailed in LNDE-RPT-20140902-024-R0. The survey was triggered by the 3-year interval set by the DMR for rich-amine service, with the previous set of readings recorded in 2011-02.

The line is classified under the rich amine interval, requiring thickness measurements every 3 years.

Thickness readings, measured 2014-09-19

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2011-02-18	8.32	8.07	0.024	0.070	>50	2017-09-19	
4-RA-101-02	8.56	3.81	2011-02-18	8.20	8.02	0.028	0.050	>50	2017-09-19	
4-RA-101-03	8.56	3.81	2011-02-18	7.98	7.88	0.039	0.028	>50	2017-09-19	
4-RA-101-04	8.56	3.81	2011-02-18	8.21	8.18	0.024	0.008	>50	2017-09-19	
4-RA-101-05	8.56	3.81	2011-02-18	8.16	7.98	0.033	0.050	>50	2017-09-19	
4-RA-101-06	8.56	3.81	—	—	7.91	—	—	—	2017-09-19	
4-RA-101-07	8.56	3.81	2011-02-18	8.30	8.22	0.021	0.022	>50	2017-09-19	
4-RA-101-08	8.56	3.81	2011-02-18	8.02	7.71	0.052	0.086	45.1	2017-09-19	
4-RA-101-09	8.56	3.81	2011-02-18	8.32	8.27	0.016	0.014	>50	2017-09-19	
4-RA-101-10	8.56	3.81	2011-02-18	8.28	8.21	0.025	0.020	>50	2017-09-19	
4-RA-101-11	8.56	3.81	2011-02-18	8.04	7.88	0.043	0.045	>50	2017-09-19	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2011-02-18	8.33	8.25	0.016	0.022	>50	2017-09-19	
4-RA-102-02	8.56	3.81	2011-02-18	7.99	7.89	0.037	0.028	>50	2017-09-19	
4-RA-102-03	8.56	3.81	2011-02-18	8.04	7.95	0.033	0.025	>50	2017-09-19	
4-RA-102-04	8.56	3.81	2011-02-18	7.96	7.93	0.033	0.008	>50	2017-09-19	
4-RA-102-05	8.56	3.81	2011-02-18	8.07	7.84	0.038	0.064	>50	2017-09-19	
4-RA-102-06	8.56	3.81	2011-02-18	8.01	7.95	0.030	0.017	>50	2017-09-19	
4-RA-102-07	8.56	3.81	2011-02-18	8.10	7.98	0.032	0.033	>50	2017-09-19	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-08	8.56	3.81	2011-02-18	7.88	7.58	0.049	0.084	45.0	2017-09-19	
4-RA-102-09	8.56	3.81	2011-02-18	7.98	7.84	0.038	0.039	>50	2017-09-19	
4-RA-102-11	8.56	3.81	2011-02-18	8.19	8.12	0.020	0.020	>50	2017-09-19	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2011-02-18	10.47	10.62	0.021	-0.042	>50	2017-09-19	
6-RA-103-02	10.97	4.57	2011-02-18	10.23	10.06	0.056	0.047	>50	2017-09-19	
6-RA-103-03	10.97	4.57	2011-02-18	10.42	10.39	0.032	0.008	>50	2017-09-19	
6-RA-103-04	10.97	4.57	2011-02-18	9.97	9.79	0.064	0.050	>50	2017-09-19	
6-RA-103-05	10.97	4.57	2011-02-18	10.10	9.81	0.062	0.081	>50	2017-09-19	
6-RA-103-06	10.97	4.57	2011-02-18	10.56	10.43	0.031	0.036	>50	2017-09-19	
6-RA-103-07	10.97	4.57	2011-02-18	10.55	10.37	0.033	0.050	>50	2017-09-19	
6-RA-103-08	10.97	4.57	2011-02-18	10.44	10.49	0.026	-0.014	>50	2017-09-19	
6-RA-103-09	10.97	4.57	2011-02-18	10.56	10.39	0.033	0.047	>50	2017-09-19	
6-RA-103-10	10.97	4.57	2011-02-18	10.05	9.93	0.059	0.033	>50	2017-09-19	
6-RA-103-11	10.97	4.57	—	—	10.10	—	—	—	2017-09-19	
6-RA-103-13	10.97	4.57	2011-02-18	10.45	10.38	0.020	0.020	>50	2017-09-19	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2011-02-18	7.95	7.84	0.036	0.031	>50	2017-09-19	
4-RA-104-02	8.56	3.81	2011-02-18	8.14	8.09	0.025	0.014	>50	2017-09-19	
4-RA-104-03	8.56	3.81	2011-02-18	8.56	8.46	0.032	0.028	>50	2017-09-19	
4-RA-104-04	8.56	3.81	2011-02-18	8.60	8.49	0.007	0.031	>50	2017-09-19	
4-RA-104-05	8.56	3.81	2011-02-18	8.57	8.37	0.015	0.056	>50	2017-09-19	
4-RA-104-06	8.56	3.81	2011-02-18	7.98	7.91	0.037	0.020	>50	2017-09-19	
4-RA-104-07	8.56	3.81	2011-02-18	7.78	7.61	0.053	0.047	>50	2017-09-19	
4-RA-104-08	8.56	3.81	2011-02-18	8.18	8.00	0.050	0.050	>50	2017-09-19	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT Level II	2014-09-19	_____